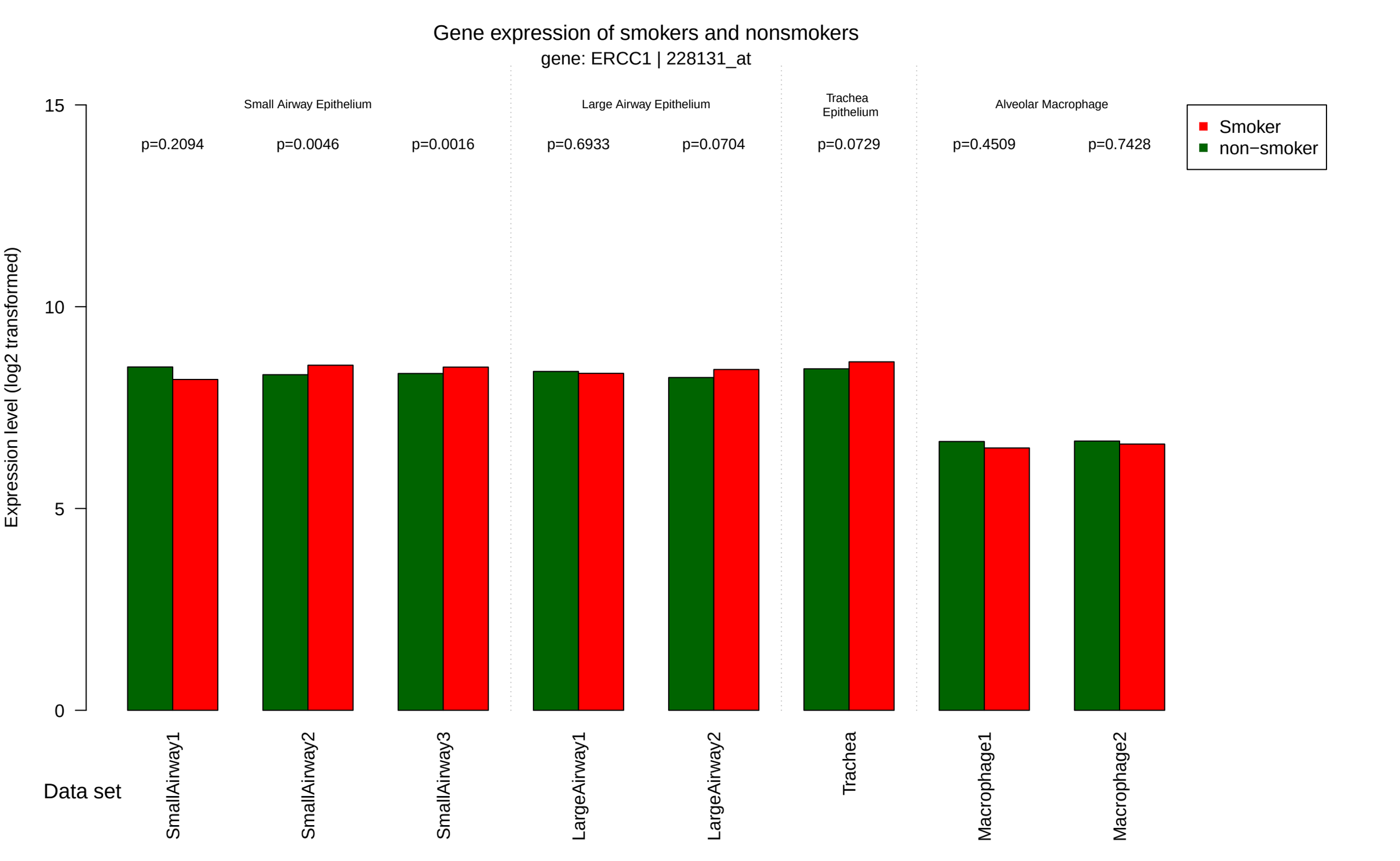
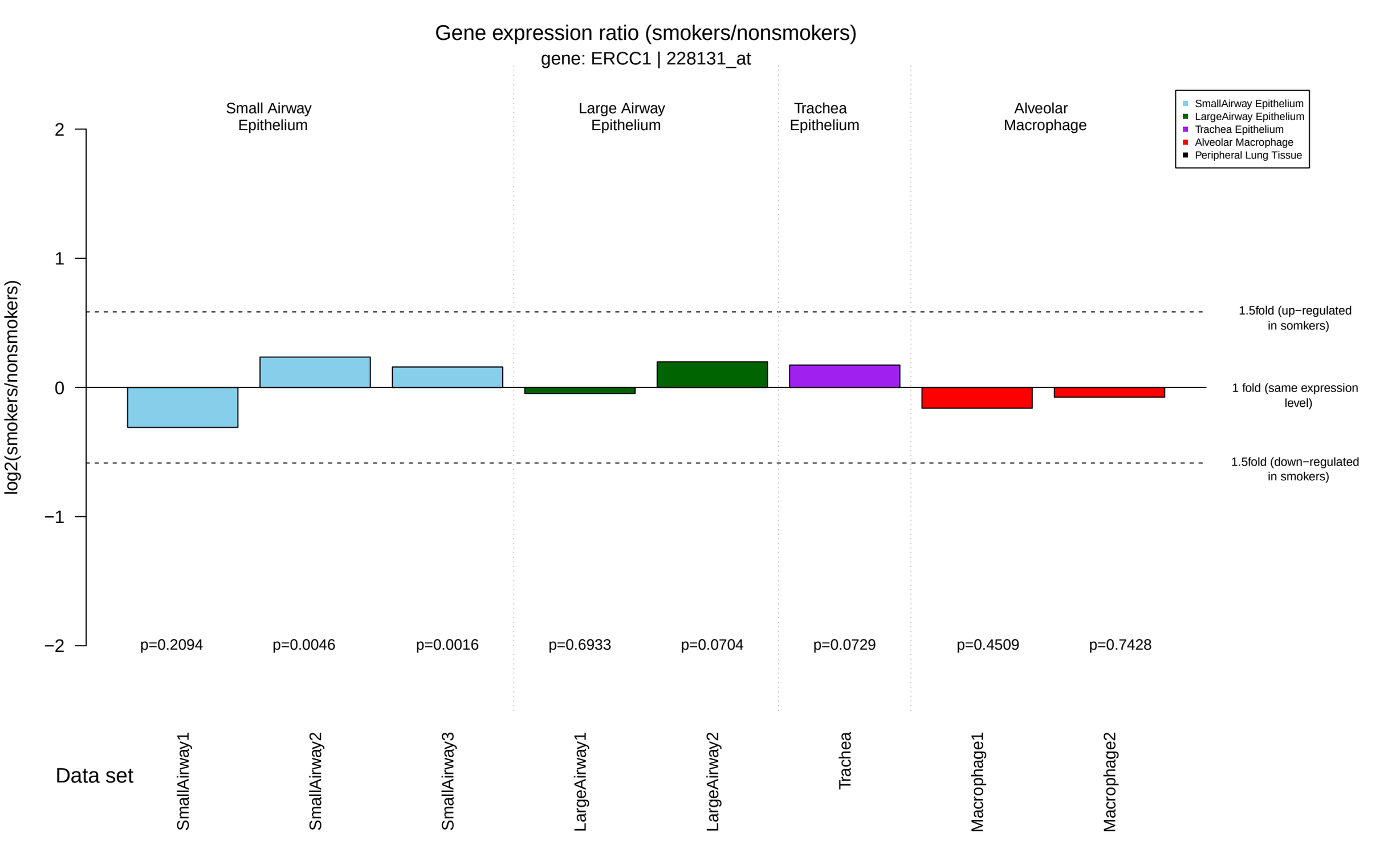
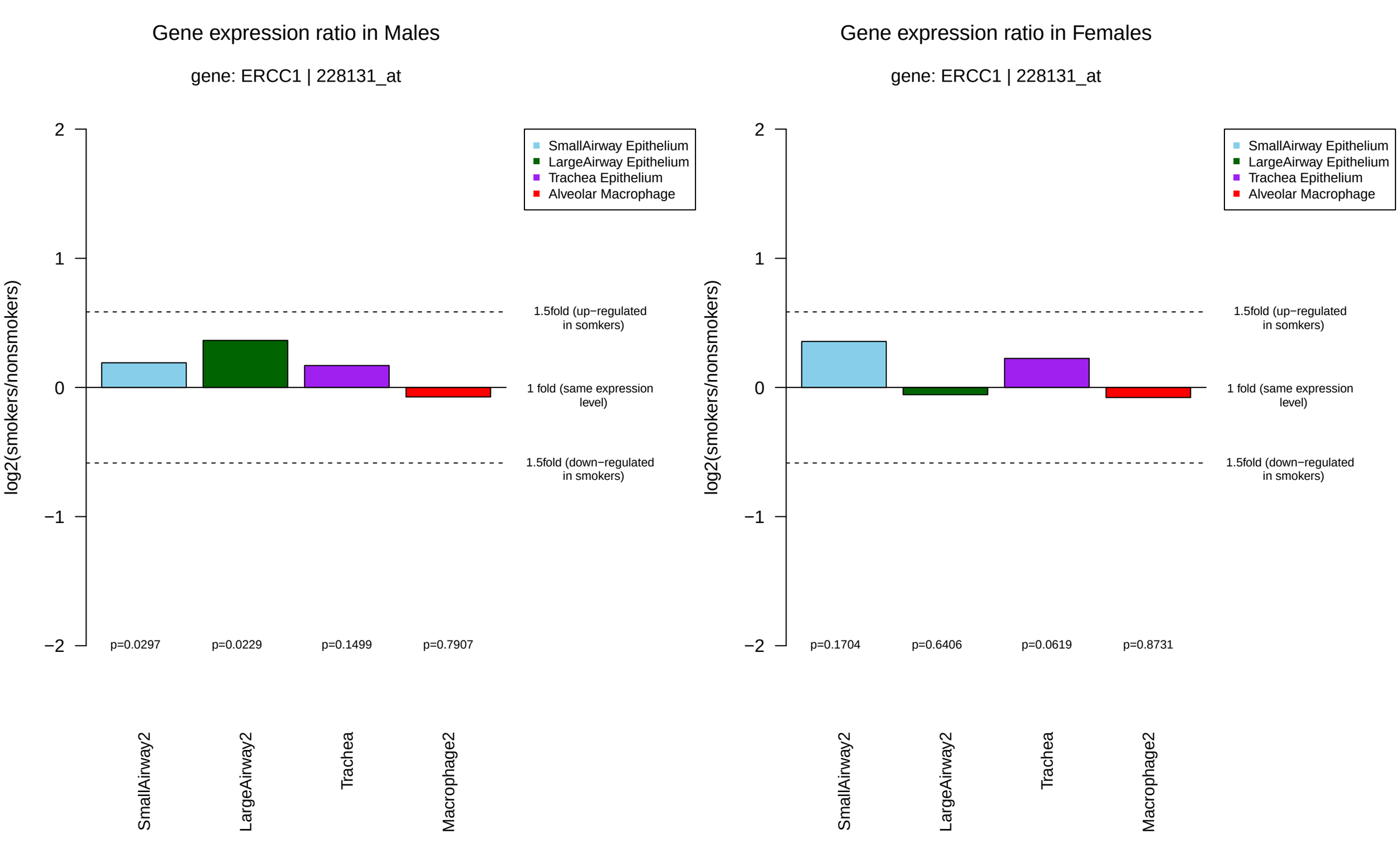
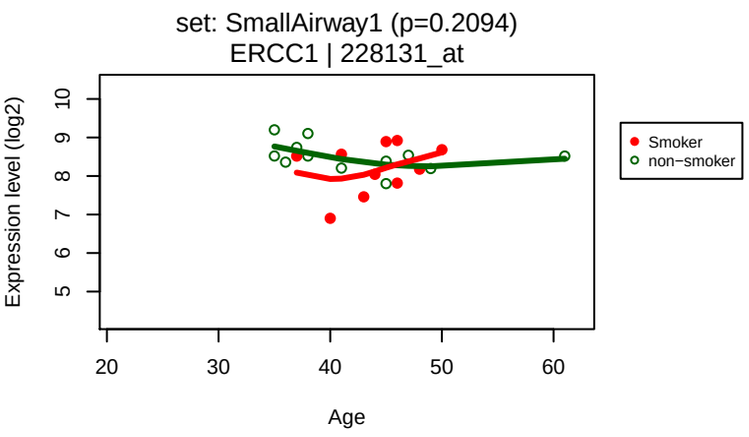
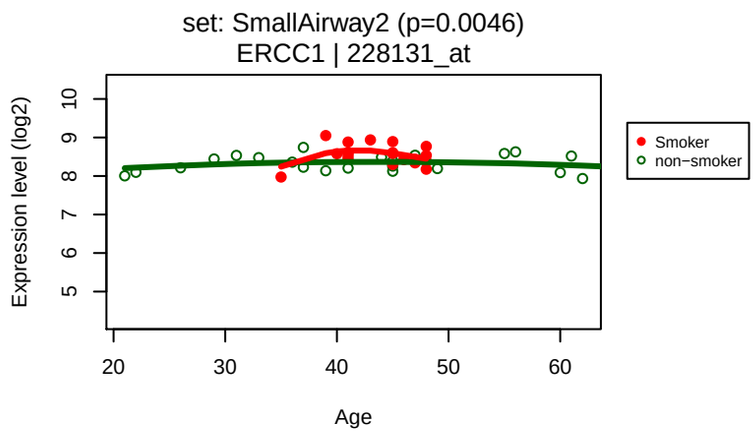
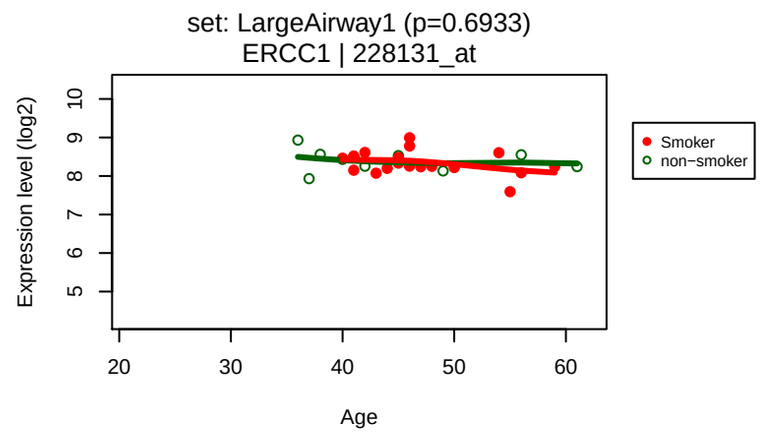

Gene expression of smokers and nonsmokers
gene: ERCC1 | 228131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2094
p=0.0046
p=0.0016
p=0.6933
p=0.0704
p=0.0729
p=0.4509
p=0.7428
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

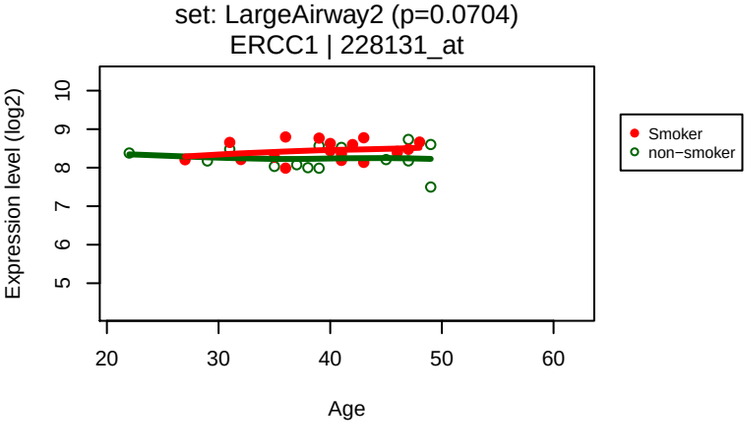

Gene expression ratio (smokers/nonsmokers)
gene: ERCC1 | 228131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2094
p=0.0046
p=0.0016
p=0.6933
p=0.0704
p=0.0729
p=0.4509
p=0.7428
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

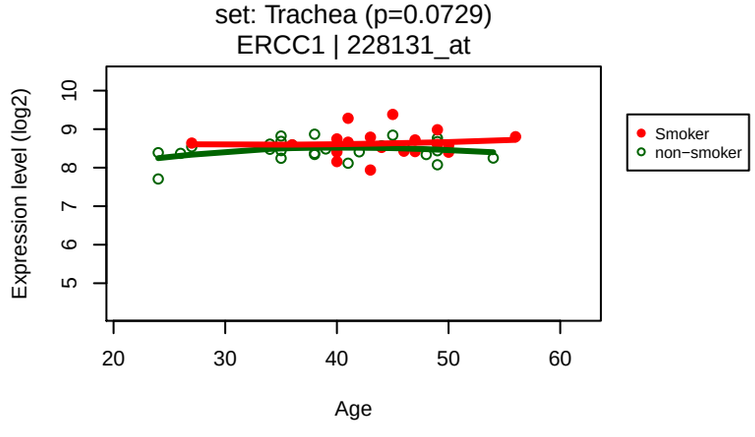

Gene expression ratio in Males
gene: ERCC1 | 228131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0297
p=0.0229
p=0.1499
p=0.7907
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERCC1 | 228131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1704
p=0.6406
p=0.0619
p=0.8731
SmallAirway2
LargeAirway2
Trachea
Macrophage2

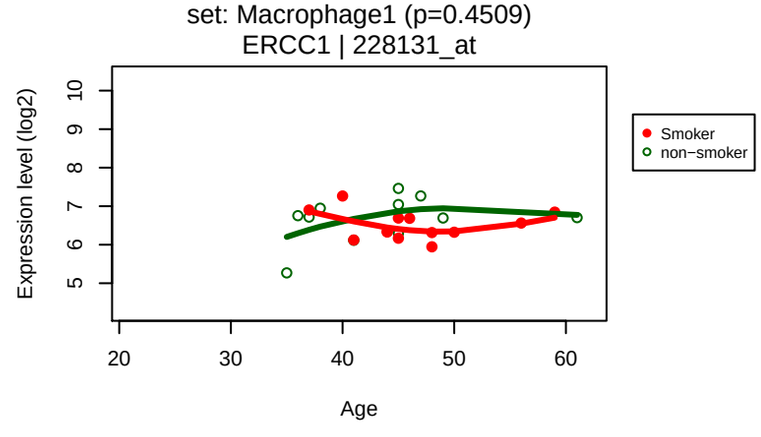

set: SmallAirway1 (p=0.2094)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker

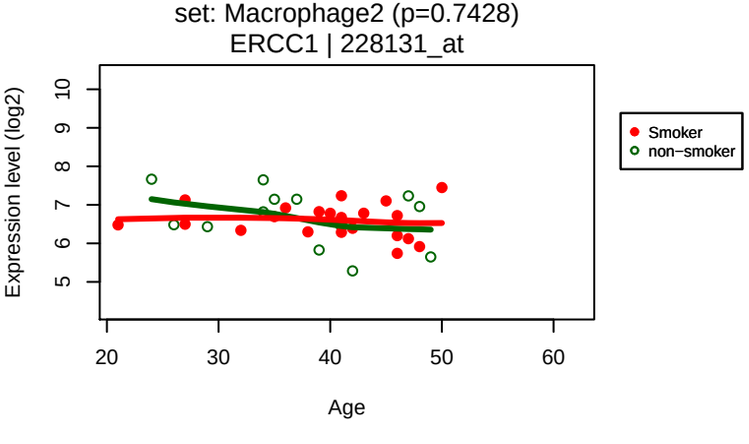

set: SmallAirway2 (p=0.0046)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker

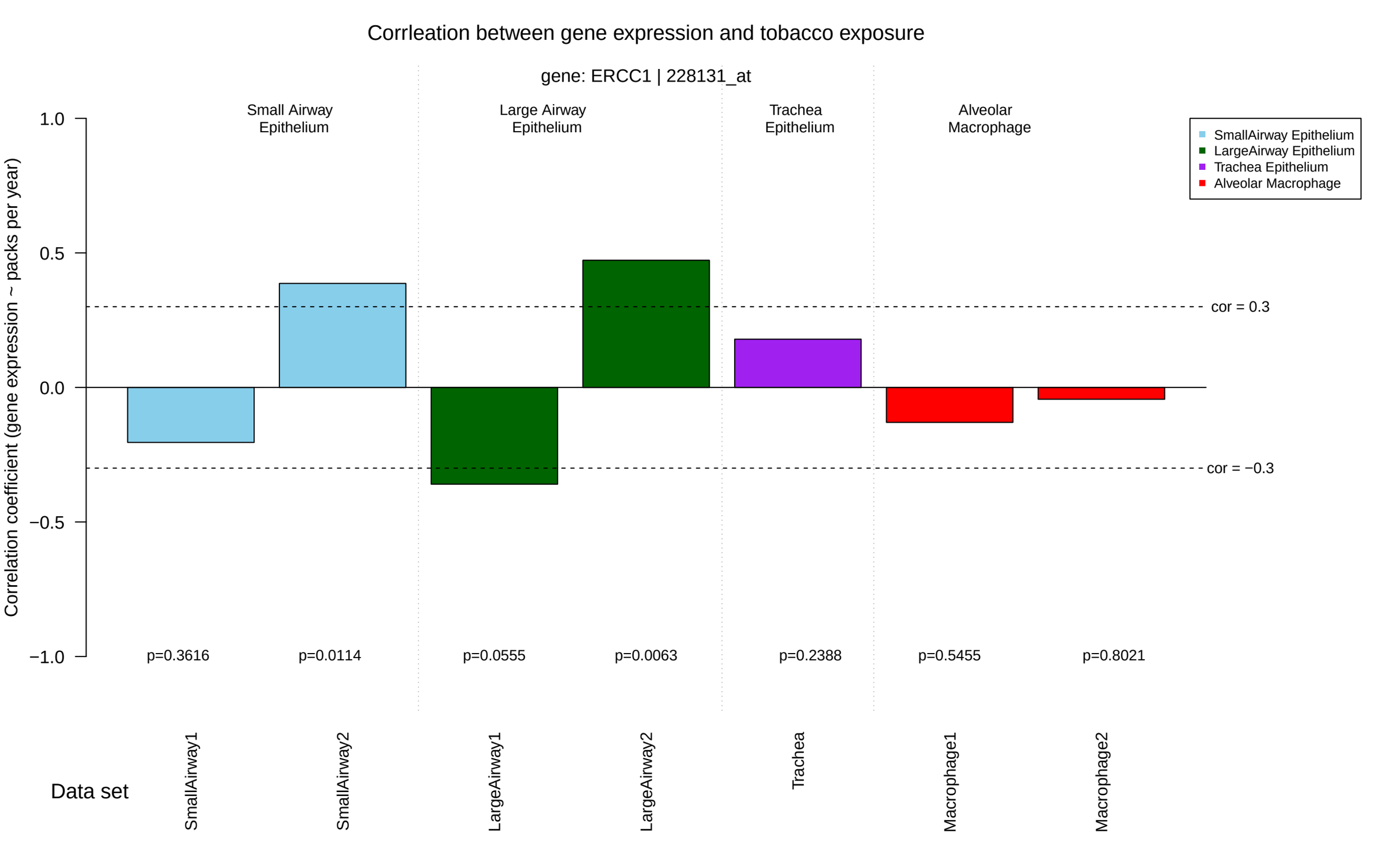

set: LargeAirway1 (p=0.6933)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0704)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0729)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4509)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7428)
ERCC1 | 228131_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERCC1 | 228131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3616
p=0.0114
p=0.0555
p=0.0063
p=0.2388
p=0.5455
p=0.8021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2